

Solution Requirements

Date	26 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

The Emotion Detection & Learning Support Engine is designed to analyze students' emotions from text and provide personalized learning support using AI. The functional requirements define the features and services the system must provide, while the non-functional requirements describe the quality attributes such as performance, security, reliability, and usability that ensure an efficient and secure user experience.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users securely access the application using a username and password.</i>	High
2	Authorization levels	<i>Student users can analyze emotions and receive recommendations, while administrators can manage system data and monitor usage.</i>	Medium
3	External interfaces	<i>Integrates with the Gemini API for AI-generated learning support and uses BiLSTM and BERT models for emotion detection.</i>	High

4	Transactions processing	<i>The system accepts text input, processes it through the emotion detection models, generates personalized recommendations, and stores the results.</i>	High
5	Reporting	<i>Displays detected emotions, confidence scores, learning suggestions, and maintains emotion history through the dashboard.</i>	Medium
6	Business rules, etc.	<i>Learning recommendations are generated based on the detected emotion, and all inputs are validated before processing.</i>	High
7	Compliance to laws or regulations	<i>User data is handled securely, API keys are protected, and privacy practices follow general data protection guidelines.</i>	Medium
8	Other	<i>The system maintains previous emotion records to support learning progress tracking.</i>	Low

Step 2: Non-Functional Requirements (NFR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)

1	Performance & Speed	<i>Emotion detection and AI response should be generated quickly.</i>	<i>Response time < 3 seconds</i>
2	Scalability	<i>The application should support multiple users simultaneously without performance degradation.</i>	<i>Supports at least 100 concurrent users</i>
3	Security & Data Privacy	<i>API keys, user inputs, and stored data must be protected from unauthorized access.</i>	<i>Secure storage using environment variables and encrypted communication</i>
4	Reliability & Availability	<i>The system should provide stable operation with minimal downtime.</i>	<i>99% uptime</i>
5	Usability & Accessibility	<i>The application should provide a simple, intuitive Streamlit interface that is easy to navigate.</i>	<i>User-friendly interface with clear navigation</i>
6	Other	<i>The codebase should be modular, maintainable, and easy to extend with additional AI models or features.</i>	<i>Well-structured and documented code</i>

